Networking Academy

CCNP Enterprise: Advanced Routing (ENARSI)

Scope and Sequence

Version 9

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Target Audience

The Cisco Networking Academy® CCNP Enterprise curriculum is designed for participants who are seeking professional-level jobs in the ICT industry or hope to fulfill prerequisites to pursue other CCNP or CCIE certifications. The entire curriculum is appropriate for learners at universities and colleges.

Prerequisites

CCNA or equivalent knowledge and skills.

Course Description

In this curriculum, Cisco Networking Academy™ participants learn, apply, and practice CCNP Enterprise knowledge and skills through a series of in-depth, hands-on experiences that reinforce their learning.

The CCNP Enterprise v9 curriculum is presented in two courses that provide integrated and comprehensive coverage of professional-level networking technologies.

Upon completion of each course, learners will be prepared to take the certification exam associated with that course.

The following two courses comprise the CCNP Enterprise v9 curriculum:

CCNP Enterprise: Core Networking v9

The Implementing and Operating Cisco Enterprise Network Core Technologies (ENCOR) course provides learners with the knowledge and skills needed to install, configure, operate, and troubleshoot an enterprise network. This course introduces overlay network design through SD-Access and SD-WAN solutions and teaches learners to implement security principles and automation and programmability within an enterprise network. This course helps prepare learners to take the 350-401 ENCOR exam, which is part of the CCIE Enterprise Infrastructure certification. This version is aligned with the updated version of the 350-401 exam, version 1.2.

Learners will be able to:

- Configure and implement identified solutions by applying planned implementation processes using Cisco IOS Software commands and applications.
- Verify appropriate show and debug commands and applications to ensure correct solution implementation and performance.
- Troubleshoot appropriate show and debug commands and applications to identify the cause of basic-level network issues and correctly implement a solution that ensures the network is performing as desired.

CCNP Enterprise: Advanced Routing v9

The Implementing Cisco Enterprise Advanced Routing and Services (ENARSI) course provides learners with the knowledge and skills needed to install, operate, manage, and troubleshoot an enterprise network. Throughout the course, learners will build the skills necessary to ensure the smooth and effective operation of these networks. Completing this course prepares them to take the Implementing Cisco Enterprise Advanced Routing and Services (300-410 ENARSI)

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exam version 1.1. Learners who pass the exam will earn the Cisco Certified Specialist - Enterprise Advanced Infrastructure Implementation certification, which is also part of the CCNP Enterprise certification.

Learners will be able to:

- Install, configure, and operate enterprise networks.
- Troubleshoot enterprise networks with a focus on advanced routing and services.
- Implement Layer 3 technologies, route manipulation, VPN technologies, network infrastructure, and network security solutions.
- Demonstrate advanced knowledge of enterprise networking tools, technologies, and concepts.

These two courses, along with the corresponding certification exams, align to the overall CCNP Enterprise certification. Both courses provide learners with extensive opportunities for hands-on practical experience and career-skills development.

Authorized Use

Access to the CCNP curriculum is granted exclusively under the terms of the Cisco Networking Academy Membership Agreement.

This content is licensed for use in academic degree programs and non-profit workforce development initiatives only.

Use of these materials for commercial training, bootcamps, or other for-profit activities outside the scope of the agreement is strictly prohibited.

Failure to adhere to these policies may lead to the termination of the Academy agreement.

Course Features

The CCNP Enterprise v9 curriculum includes many features to engage and assess learner knowledge and skills.

Curriculum

The online curriculum has been repurposed for use by the Cisco Networking Academy from the premium offerings at Cisco U.

The Cisco U. built-in, hands-on labs, (using the terminology “Discovery”), available to Cisco U. premium subscribers, have been stripped and are only available at Cisco Networking Academy as video-based demonstrations.

The course at Cisco Networking Academy has been expanded with additional hands-on practice activities that leverage the Cisco Hardware Classroom Equipment, Cisco Packet Tracer, or Cisco Modeling Labs free edition (CML-Free). These lab activities are threaded throughout the course using the Assignment Flow.

Assessments

Assessments and practice activities are focused on specific certification competencies to increase knowledge and skill retention.

Embedded module-group assessments provide immediate feedback to support the evaluation of knowledge and acquired skills.

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Activities

Hands-on labs help students develop critical thinking and complex, problem-solving skills.

It is recommended to leverage CML-Free to emulate real network operating systems and practice important network virtualization skills and concepts, such as a digital twin.

Cisco Packet Tracer simulation-based learning activities are added as optional instructor resources to support vital CCNA skills.

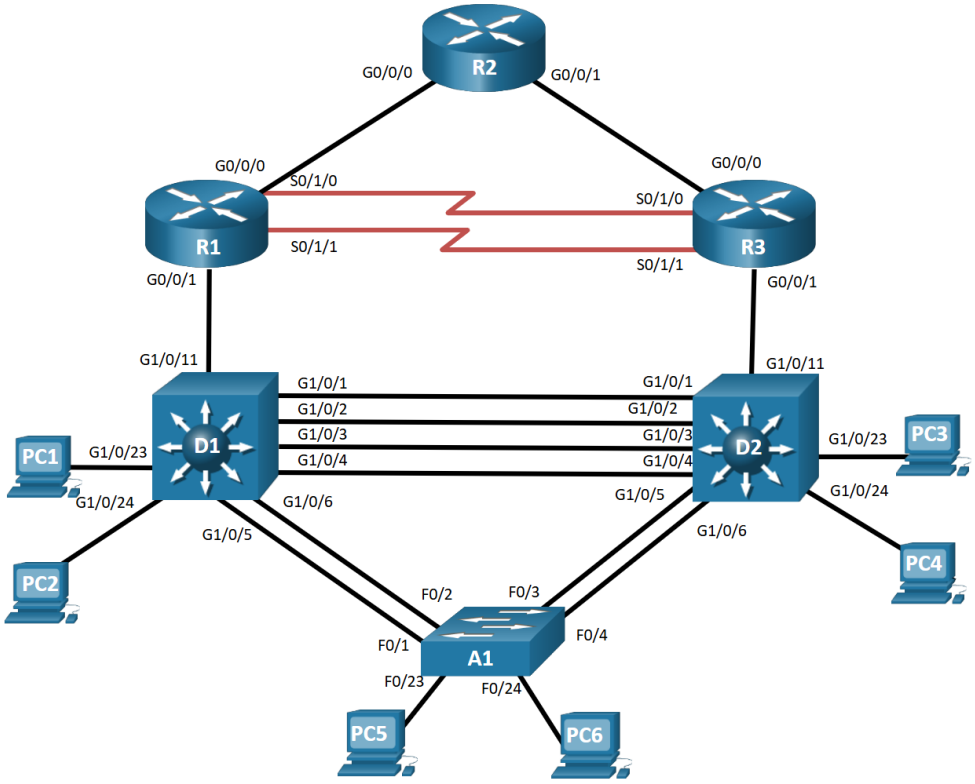
Video demonstrations give insight into complex workflows and processes providing exposure to advanced networking tools and operations.

Equipment Requirements

Detailed equipment information, including descriptions and part numbers, is available in the CCNP Enterprise Equipment List, which is located on the Cisco NetAcad Equipment Information site (https://www.netacad.com/resources/program-info/equipment-information).

The labs in the CCNP Enterprise curriculum use the following topology and equipment.

CCNP Enterprise Baseline Physical Topology (for both ENCOR and ENARSI)



Baseline Equipment Bundle

- 3x C8200L-1N-4T (2 with NIM-2T)
- 2x C9300L-24T-4G-A
- 1x C9200L-24T-4G-E
- Ethernet cables as shown in the topology
- 2x CAB-SS-V35MT= (10' DTE Serial Cable)

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- 2x CAB-SS-V35FC= (10' DCE Serial Cable)
- PCs - minimum system requirements
    - CPU: Intel Core i5 (8th Gen or newer) / AMD Ryzen 5 or equivalent
        - Must support hardware virtualization (Intel VT-x or AMD-V) for lab simulation software
    - Operating System, such as Microsoft Windows, Linux, or Mac OS
    - RAM: 16 GB (Recommended for running CML/Virtual Machines); 8 GB (Minimum)
    - Storage: 20 GB of free disk space
    - Ethernet port (RJ-45) or USB Ethernet adapter (for connecting to physical lab equipment)
    - Display resolution: 1920 x 1080 (FHD) or higher
    - Latest video card drivers, operating system updates, and a modern web browser (Chrome, Firefox, or Edge)
- Internet connection for lab and study PCs

Software

- Cisco IOS versions:
    - Routers: Version IOSXE-17.15.4c or higher.
    - Layer 3 Switches: Version IOS-XE 17.12.6 or higher.
    - Layer 2 Switches: Version IOS 17.12.6 or higher
- Packet Tracer v9 or newer
- Open-source server software for various services and protocols, such as HTTP, DHCP, FTP, TFTP, etc.
- Terminal emulation and SSH client software, such as Tera Term and PuTTY for lab PCs
- Oracle VirtualBox, latest version
- VMware Workstation, latest version
- Wireshark version: latest stable version
- Terminal emulation software for the installed PC operating system

Course Outline

The CCNP Enterprise: Advanced Routing course is structured into 6 domains (module groups) that cover specific competencies. The table below details the modules and their associated objectives. Each module is an integrated unit of learning that consists of content, activities, and assessments that target a specific set of competencies. The size of the module depends on the depth of knowledge and skill needed to master the competencies.

No.	Module Name	Objective
Layer 3 Technologies - EIGRP		
1	Implementing EIGRP	Implement EIGRP features and operations, differentiate between Classic and Named modes, and understand routing exchange and feasibility conditions.
2	Optimizing EIGRP	Optimize EIGRP performance by configuring queries, stub routers, Stuck in Active (SIA) mitigation, and summary routes.
3	Troubleshooting EIGRP	Troubleshoot common EIGRP issues regarding neighbor formation, routing tables, stub configurations, and summarization.
Layer 3 Technologies - OSPF		
4	Implementing OSPF	Implement OSPF features, hierarchical structures, and message types, and analyze the Link State Database (LSDP) synchronization and SPF algorithm.
5	Optimizing OSPF	Optimize OSPF using route summarization, special area types (Stubby Areas), default routes, and virtual links.

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6	Troubleshooting OSPF	Troubleshoot OSPF adjacency, routing issues, path selection, special area configurations, and summarization.
	Routing Manipulation	
7	Configuring Redistribution	Configure route redistribution between protocols using default metrics, mutual redistribution, and manipulation tools like route maps and distribute lists.
8	Troubleshooting Redistribution	Troubleshoot route redistribution issues, including metric mismatches and route feedback loops.
9	Implementing Path Control	Implement path control mechanisms using Policy-Based Routing (PBR), IP-SLA, and Bidirectional Forwarding Detection (BFD).
	Layer 3 Technologies - BGP	
10	Implementing IBGP	Implement Internal BGP (IBGP) and External BGP (EBGP), covering fundamentals, neighbor relationships, path attributes, and transit AS functionality.
11	Optimizing BGP	Optimize BGP using attributes (Weight, Local Preference, MED, AS-Path), route filtering, peer groups, and Route Reflectors.
12	Implementing MP-BGP	Configure Multiprotocol BGP (MP-BGP) to support and filter both IPv4 and IPv6 address families.
13	Troubleshooting BGP	Troubleshoot BGP neighbor relationships, IBGP configurations, and MP-BGP issues.
	MPLS and VPN Technologies	
14	Exploring MPLS	Explain MPLS concepts, terminology, Label Switching Router (LSR) types, and traffic forwarding mechanisms compared to traditional IP routing.
15	Introducing MPLS L3 VPN Architecture	Describe MPLS Layer 3 VPN architecture, including PE routers, VRFs, Route Distinguishers (RD), Route Targets (RT), and packet flow.
16	Introducing MPLS L3 VPN Routing	Explain MPLS Layer 3 VPN routing requirements, table operations, and the end-to-end flow of routing updates and packet forwarding.
17	Configuring VRF-Lite	Configure and verify VRF-Lite, including VRF-aware routing protocols and VRF table management.
18	Implementing DMVPN	Implement DMVPN using GRE, mGRE, and NHRP, covering hub-and-spoke configurations and authentication.
	Infrastructure Security and Services	
19	Implementing DHCP	Implement DHCP for IPv4 and IPv6, including relays, address bindings, and SLAAC (stateful and stateless).
20	Introducing IPv6 First Hop Security	Secure IPv6 networks using First Hop Security tools such as ND inspection, RA Guard, Source Guard, and Destination Guard.
21	Securing Cisco Routers	Secure Cisco routers using Access Control Lists (ACLs), Control Plane Policing (CoPP), and Unicast Reverse Path Forwarding (uRFP).
22	Troubleshooting Infrastructure Security and Services	Troubleshoot infrastructure security services including AAA, SNMP, Syslog, and NetFlow.
23	Troubleshooting with DNA Center Assurance	Monitor and troubleshoot networks using Cisco DNA Center Assurance tools, AI Network Analytics, and Path Trace.

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